

Coequivalence relations and descent in modal logic

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Coequivalence relations

Definability

We are interested in the following family of problems in logic:

What **implicitly defined** objects can a logic **explicitly define**?

Beth definability property: can implicit functions be made explicit?

Definition

A logic L has the **Beth definability property** if for any $\varphi(\bar{p}, q)$:

$$\text{if } \varphi(\bar{p}, q) \wedge \varphi(\bar{p}, q') \vdash_L q \leftrightarrow q' ,$$

then there exists $\psi(\bar{p})$ s.t. $\varphi(\bar{p}, q) \vdash_L q \leftrightarrow \psi(\bar{p})$.

Uniform interpolation: explicitly defining **propositional quantifiers**.

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A definability problem first studied by Ghilardi and Zawadowski (2002):

Given a “provable equivalence relation” of the logic, can the logic explicitly define the **equivalence classes**?

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Given $\tau(\bar{p})$, a τ -coequivalence relation is $\rho(\bar{p}_1, \bar{p}_2)$ such that:

- (Context) $\rho(\bar{p}_1, \bar{p}_2) \vdash_L \tau(\bar{p}_1) \wedge \tau(\bar{p}_2)$;
- (Reflexivity) $\tau(\bar{p}) \vdash_L \rho(\bar{p}, \bar{p})$;
- (Symmetry) $\rho(\bar{p}_1, \bar{p}_2) \vdash_L \rho(\bar{p}_2, \bar{p}_1)$;
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Separation of variables

Definition

A τ -coequivalence relation $\rho(\bar{p}_1, \bar{p}_2)$ is said to **separate variables** if there are $\psi_1(\bar{p}), \dots, \psi_n(\bar{p})$ such that

$$\rho(\bar{p}_1, \bar{p}_2) \dashv\vdash_L \tau(\bar{p}_1) \wedge \tau(\bar{p}_2) \wedge \bigwedge_{i=1}^n (\psi_i(\bar{p}_1) \leftrightarrow \psi_i(\bar{p}_2)) .$$

Definition

We say that a logic L has the **coequivalence separation property (CoSP)** if every coequivalence relation separates variables.

Model theoretic interpretation: an instance of **elimination of imaginaries**¹.

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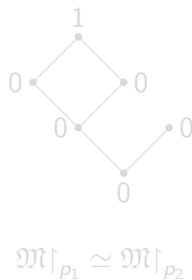
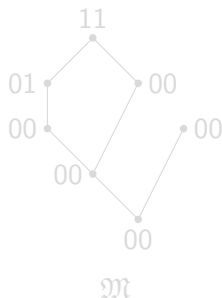
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Failure of the CoSP

CPC enjoys the CoSP: a model w over $\bar{p}_1$ and $\bar{p}_2$ is **uniquely determined** by the restrictions $w|_{\bar{p}_1}$ and $w|_{\bar{p}_2}$.²

The CoSP fails in IPC [GZ2002]: $\rho(p_1, p_2) := \neg\neg(p_1 \wedge p_2) \vee (p_1 \leftrightarrow p_2)$



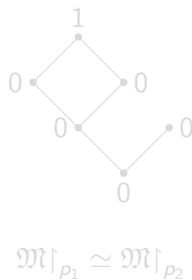
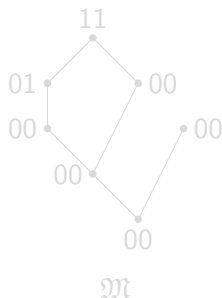
The CoSP fails also in KC, K, K4, S4.

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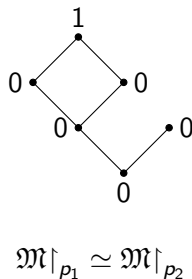
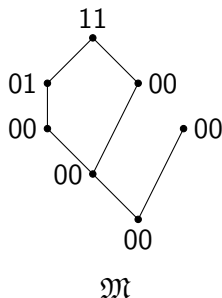
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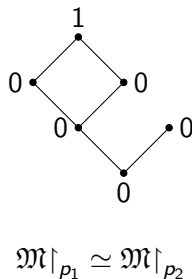
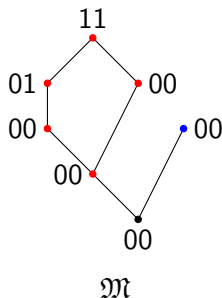
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Categorical interpretation

$\mathbf{C}$ with finite limits. $X \otimes Y$ product of X and Y in $\mathbf{C}$.

Definition

An **equivalence relation** over X is a mono $(p_1, p_2): R \longrightarrow X \otimes X$:

- (R) $r: X \longrightarrow R$ s.t. $p_1 r = \text{id}_X$ and $p_2 r = \text{id}_X$;
- (S) $s: R \longrightarrow R$ s.t. $p_1 s = p_2$ and $p_2 s = p_1$;
- (T) $t: R \otimes_X R \longrightarrow R$ s.t. $p_1 t = p_1 q_1$ and $p_2 t = p_2 q_2$:

$$\begin{array}{ccc} R \otimes_X R & \xrightarrow{q_2} & R \\ q_1 \downarrow & \lrcorner & \downarrow p_1 \\ R & \xrightarrow{p_2} & X \end{array}$$

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Effective equivalence relations

Effectiveness: “pick equivalence classes with a map”.

Definition

An equivalence relation $(p_1, p_2): R \longrightarrow X \otimes X$ is said to be **effective** if there is $f: X \longrightarrow Y$ such that

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ p_1 \downarrow & \lrcorner & \downarrow f \\ X & \xrightarrow{f} & Y \end{array}$$

For normal modal logics (or superintuitionistic logics), then:

Theorem

L has the CoSP iff every equivalence relation in $\text{Alg}(L)_{\text{fp}}^{\text{op}}$ is effective.^a

^aTogether with regularity, this property is called Barr-exactness.

Coequivalence separation in S5

The S5 case

S5-algebras are Boolean algebras A , with $\Box: A \rightarrow A$ s.t. $(\Diamond := \sim \Box \sim)$

$$\Box 1 = 1; \quad \Box(a \wedge b) = \Box a \wedge \Box b; \quad \Box a \leq a;$$

$$\Box a \leq \Box \Box a; \quad \Diamond a \leq \Box \Diamond a.$$

We have $\text{Alg}(\text{S5})_{\text{fp}}^{\text{op}}$:⁴

- **Objects:** finite families $\mathcal{X}$ of non-empty finite sets;
- **Arrows** $\mathcal{X} \rightarrow \mathcal{Y}$: functions $f: \mathcal{X} \rightarrow \mathcal{Y}$ with a family of **surjective** functions $f_X: X \rightarrow f(X)$, for each $X \in \mathcal{X}$.

⁴This representation goes back to the work of Horn (1978) and Bass (1958).

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Definition

A **bisimulation** from $X \neq \emptyset$ to $Y \neq \emptyset$ is $A \subseteq X \times Y$:

- $\forall x \in X, \exists y \in Y$ s.t. $(x, y) \in A$;
- $\forall y \in Y, \exists x \in X$ s.t. $(x, y) \in A$.

$\text{Bis}(X, Y)$ is the set of bisimulations from X to Y .

Definition

For $\mathcal{X} \in \text{Alg}(\text{S5})_{\text{fp}}^{\text{op}}$,

$$\text{Bis}(\mathcal{X}) := \coprod_{X_1, X_2 \in \mathcal{X}} \text{Bis}(X_1, X_2)$$

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Equivalence relations in $\text{Alg}(S5)_{\text{fp}}^{\text{op}}$.

Definition

$\mathcal{R} \subseteq \text{Bis}(\mathcal{X})$ is **equivalential** if:

- (Identity) for $X \in \mathcal{X}$,

$$\Delta_X := \{(x, x) \mid x \in X\} \in \mathcal{R} ;$$

- (Dagger) for $X_1, X_2 \in \mathcal{X}$ and $A \in \text{Bis}(X_1, X_2)$,

$$A \in \mathcal{R} \implies A^\dagger := \{(x_2, x_1) \mid (x_1, x_2) \in A\} \in \mathcal{R} ;$$

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$$A, B \in \mathcal{R} \implies \text{Comp}(P) := \{(x_1, x_3) \mid ((x_1, x_2), (x_2, x_3)) \in P\} \in \mathcal{R} .$$

^a $((x_1, x_2), (x'_2, x_3)) \in P \implies x_2 = x'_2$.

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Failure of CoSP in S5

Proposition

Equivalential $\mathcal{R} \subseteq \text{Bis}(\mathcal{X})$ correspond to coequivalence relations for S5.

$\mathcal{R}$ separates variables iff, for each $X_1, X_2 \in \mathcal{X}$ and $A, B \in \text{Bis}(X_1, X_2)$,

- (D) $A \in \mathcal{R}$ and $B \subseteq A \implies B \in \mathcal{R}$;*
- (U) $A, B \in \mathcal{R} \implies A \cup B \in \mathcal{R}$.*

Theorem [R. N. Almeida, M. D.]

The CoSP fails in S5.

Proof. Take $X = \{1, 2\}$ and $\mathcal{X} = \{X\}$. Consider

$$A := \{(1, 2), (2, 1)\} \in \text{Bis}(X, X) .$$

$\mathcal{R} := \{\Delta_X, A\} \subseteq \text{Bis}(\mathcal{X})$ is equivalential, but $A \cup \Delta_X = X \times X \notin \mathcal{R}$. □

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$\mathcal{R}$ separates variables iff, for each $X_1, X_2 \in \mathcal{X}$ and $A, B \in \text{Bis}(X_1, X_2)$,

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Theorem [R. N. Almeida, M. D.]

The CoSP fails in S5.

Proof. Take $X = \{1, 2\}$ and $\mathcal{X} = \{X\}$. Consider

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$\mathcal{R} := \{\Delta_X, A\} \subseteq \text{Bis}(\mathcal{X})$ is equivalential, but $A \cup \Delta_X = X \times X \notin \mathcal{R}$. □

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Descent and local coequivalences

Due to the failure of CoSP in IPC, Ghilardi and Zawadowski (2002):

Does IPC satisfy a weaker definability property concerning “**separation of two sets of independent variables**”?⁵

⁵As the CoSP is related to Barr-exactness, they suggested this was related to “all regular epis being of effective descent” (implied by, but not implying, Barr-exactness).

Local transition terms and coequivalence relations

Let $\beta(\bar{p}, \bar{q}) \vdash_L \alpha(\bar{p})$ and $\gamma(\bar{p}, \bar{q}, \bar{r}) \vdash_L \beta(\bar{p}, \bar{q})$.

Definition

$(\xi_r(\bar{p}, \bar{q}_1, \bar{q}_2, \bar{r}) : r \in \bar{r})$, is a sequence of **local transition terms** if:

- (γ -compatibility): $\gamma(\bar{p}, \bar{q}_1, \bar{r}) \wedge \beta(\bar{p}, \bar{q}_2) \vdash_L \gamma(\bar{p}, \bar{q}_2, \bar{\xi})$;
- (Identity): for each $r \in \bar{r}$, $\gamma(\bar{p}, \bar{q}, \bar{r}) \vdash_L r \leftrightarrow \xi_r(\bar{p}, \bar{q}, \bar{q}, \bar{r})$;
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Intuition: $\bar{q}$ and $\bar{r}$ are sets of independent variables; **the local transition moves $\bar{r}$** , consistently with γ , from some $\bar{q}_1$ to some $\bar{q}_2$.

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The **local γ -coequivalence relation** associated is

$$\gamma(\bar{p}_1, \bar{q}_1, \bar{r}_1) \wedge \gamma(\bar{p}_2, \bar{q}_2, \bar{r}_2) \wedge \bigwedge_{p \in \bar{p}} (p_1 \leftrightarrow p_2) \wedge \bigwedge_{r \in \bar{r}} (\xi_r(\bar{p}_1, \bar{q}_1, \bar{q}_2, \bar{r}_1) \leftrightarrow r_2).$$

Intuition: local coequivalence relations indentify variables that are **in the same orbit of the local transition**.

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We say that L has the **local coequivalence separation property (LCoSP)** if every local coequivalence relation separates variables.

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Slice categories and pullbacks

$\mathbf{C}$ with pullbacks, $p: E \longrightarrow B$ in $\mathbf{C}$.

Definition

For $B \in \mathbf{C}$, the **slice category** over B is $\mathbf{C}/B$:

- **Objects:** pairs (Y, g) , with $g: Y \longrightarrow B$ in $\mathbf{C}$;
- **Arrows** $(Y, g) \longrightarrow (Y', g')$: arrows $h: Y \longrightarrow Y'$ in $\mathbf{C}$ s.t. $g'h = g$.

There is an adjunction $p_! \dashv p^*: \mathbf{C}/B \longrightarrow \mathbf{C}/E$ s.t.

- p^* sends $(Y, g) \in \mathbf{C}/B$ to $(E \otimes_B Y, \pi_1) \in \mathbf{C}/E$, where

$$\begin{array}{ccc} E \otimes_B Y & \xrightarrow{\pi_2} & Y \\ \pi_1 \downarrow & \lrcorner & \downarrow g \\ E & \xrightarrow{p} & B \end{array}$$

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The **category of descent data** $\text{Des}(p)$ is the category of algebras for the monad $p^*p_!$ on $\mathbf{C}/E$.

An object of $\text{Des}(p)$ is $(X, f; \xi)$, where

- $(X, f) \in \mathbf{C}/E$;
- $\xi: E \otimes_B X \rightarrow X$ in $\mathbf{C}$ s.t.

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Effective descent morphisms

Effective descent morphisms $p: E \longrightarrow B$ in $\mathbf{C}$ are those morphisms which facilitate an **algebraic description** of $\mathbf{C}/B$ by means of $\mathbf{C}/E$.

Definition

A morphism $p: E \longrightarrow B$ is of **effective descent** if the comparison functor $\Phi^p: \mathbf{C}/B \longrightarrow \text{Des}(p)$ is an equivalence.

Effective descent morphisms are always **regular epis**.

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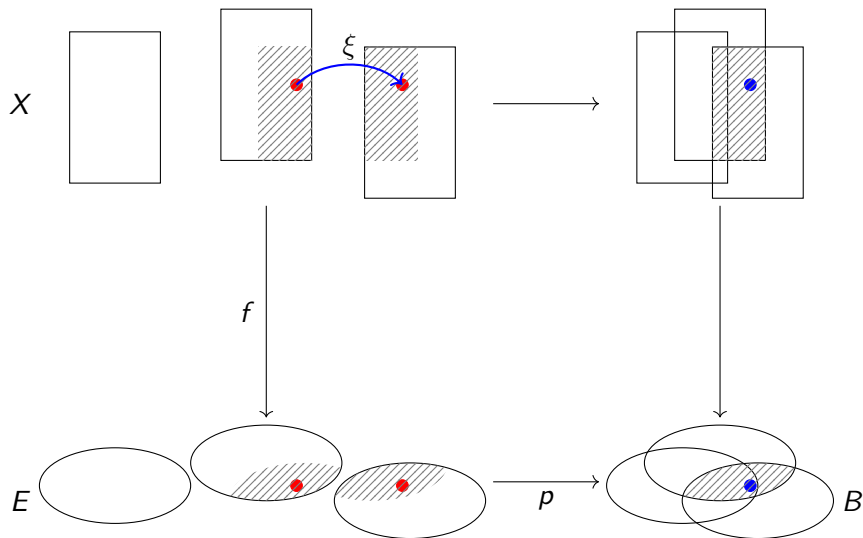
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Descent pictured



For normal modal logics (or superintuitionistic logics), then:

Theorem

L has the LCoSP iff every regular epi in $\text{Alg}(L)_{\text{fp}}^{\text{op}}$ is of effective descent.^a

^aTogether with regularity, this property is called almost Barr-exactness.

Compatible bisimulations

Let us go back to $\text{Alg}(S5)_{\text{fp}}^{\text{op}}$ and generalize $\text{Bis}(\mathcal{X})$.

Definition

For $q: \mathcal{X} \longrightarrow \mathcal{Z}$ and $p: \mathcal{Y} \longrightarrow \mathcal{Z}$ in $\text{Alg}(S5)_{\text{fp}}^{\text{op}}$,

$$\text{Bis}(q, p) := \coprod_{\substack{X \in \mathcal{X}, Y \in \mathcal{Y} \\ q(X) = p(Y)}} \{A \in \text{Bis}(X, Y) \mid (\forall (x, y) \in A) (q_X(x) = p_Y(y))\}$$

is the family of (q, p) -compatible bisimulations.

S5 enjoys the LCoSP

Proposition

$\mathcal{R} \subseteq \text{Bis}(\mathcal{X})$ correspond to a local coequivalence relation for S5 iff there are $f: \mathcal{X} \rightarrow \mathcal{Y}$ and $p: \mathcal{Y} \rightarrow \mathcal{Z}$ in $\text{Alg}(\text{S5})_{\text{fp}}^{\text{op}}$ s.t.

- (Compatibility) for each $A \in \mathcal{R}$,

$$A_f := \{(x_1, f_{X_2}(x_2)) \mid (x_1, x_2) \in A\} \in \text{Bis}(pf, p)$$

and $A \rightarrow A_f$ is bijective;

- (Lift) for each $D \in \text{Bis}(pf, p)$, there is a unique $A \in \mathcal{R}$ s.t. $A_f = D$.

Theorem [R. N. Almeida, M. D.]

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CoSP and LCoSP natural in the study of propositional non-classical logics, from a categorical, model theoretic and geometric point of view.

Future work:

- systematic approach to the study of CoSP and LCoSP;
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Thanks for your attention!

CPC enjoys the CoSP

Let $\rho(\bar{p}_1, \bar{p}_2)$ be a $\top$ -coequivalence relation. Valuations $v: \bar{p} \rightarrow \{0, 1\}$:

$$v \sim_\rho v' \iff \exists v_0: \bar{p}_1, \bar{p}_2 \rightarrow \{0, 1\} \text{ s.t. } \begin{cases} v_0 \upharpoonright_{\bar{p}_1} = v \\ v_0 \upharpoonright_{\bar{p}_2} = v' \\ v_0 \Vdash \rho. \end{cases}$$

$C_1, \dots, C_n$ equivalence classes, $\psi_i(\bar{p})$ in DNF: $v \Vdash \psi_i \iff v \in C_i$.

For $w: \bar{p}_1, \bar{p}_2 \rightarrow \{0, 1\}$,

$$w \Vdash \bigwedge_{i=1}^n (\psi_i(\bar{p}_1) \leftrightarrow \psi_i(\bar{p}_2)) \iff w \upharpoonright_{\bar{p}_1} \sim_\rho w \upharpoonright_{\bar{p}_2}$$
$$\exists v_0 \text{ s.t. } \begin{cases} v_0 \upharpoonright_{\bar{p}_1} = w \upharpoonright_{\bar{p}_1} \\ v_0 \upharpoonright_{\bar{p}_2} = w \upharpoonright_{\bar{p}_2} \\ v_0 \Vdash \rho \end{cases} \iff w = v_0 \Vdash \rho.$$